



OREGON REAL ESTATE VALUATION & MARKET INTELLIGENCE

Exploratory Analysis, Data Transformation, and Executive Presentation Dashboard

Author: Mafeng Daniel | Tool: Microsoft Excel (Power Query, Pivot Tables, Interactive Dashboard)

Executive Project Overview

High-level summary of the Oregon 2026 residential and commercial property market

TOTAL LISTING PORTFOLIO

\$6.78 Billion

Cumulative valuation across 9,999 listings

ACTIVE PROPERTIES

9,999 Units

Comprehensive statewide Oregon coverage

AVERAGE MARKET PRICE

\$678,467

Median driven by strong single family inventory

AVERAGE PRICE / SQFT

\$365.74

Benchmark metric across finished living space

Commercial Analytics Objective

This end-to-end commercial intelligence project ingests raw, unstandardized statewide property records and converts them into an executive decision model. The workflow addresses ten distinct strategic questions formulated by institutional investors and regional residential developers.

Key Deliverables Established in Excel:

1. Data Cleansing: Normalization of missing values, string case inconsistencies, and non-standardized garage flags.
2. Metric Engineering: Derivation of Price per Square Foot and Era/Decade categorization for temporal trend analysis.
3. Dynamic Pivot Engine: 10 structured Pivot Tables isolating unit counts, total volumes, and multi-variable averages.
4. Presentation Dashboard: Fully linked interactive sheet featuring 10 charts, KPI blocks, and interactive slicers.

Data Cleansing and Pipeline Engineering

Transforming 10,000 raw records into institutional-grade analytical tables

BEFORE: Raw Unprocessed Dataset ('dirty Data')

- Record Count: 10,000 rows across 12 unvalidated attributes.
- Inconsistent Categorization: 'single_family' snake_case mixed with title casing.
- Missing Numerical Values: Unhandled nulls across garage capacity, stories, and bathroom configurations.
- Unstandardized Garage Encoding: Numeric floating-point decimals (e.g., 2.0) and missing values instead of clean boolean status.
- No Temporal Grouping: Build years spanned from 1852 to 2027 without decade aggregation.
- Missing Pricing Efficiency Metric: No calculated Price per Square Foot ratio.

AFTER: Standardized Model (`'oregon_real_estate_2026_ultimat'`)

- Uniform Property Taxonomies: Standardized to 8 clean categories ('Single family', 'Land', 'Condos', 'Townhome', 'Multi family', 'Farm').
- Boolean Garage Normalization: Mapped to binary 'Yes' (6,441 units) and 'No' (3,558 units).
- Decade Segmentation Engine: Automated decade grouping formula classifying inventory into 19 distinct construction eras.
- Price per Square Foot Ratio: Calculated formula ($\text{listPrice} / \text{sqft}$) revealing underlying unit economics.
- Mathematical Boundary Validation: Cleaned null-division errors across land records where finished sqft is not applicable.
- Full Audit Trail: Preserved original raw data tab to maintain complete data lineage and transformation transparency.

Property Distribution and Market Composition

Analysis of 9,999 statewide listings by asset class and volume concentration

Property Type	Units	Market Share	Average Price	Total Volume
Single family	6,383	63.8%	\$745,355	\$4.76 Billion
Land / Acreage	1,714	17.1%	\$335,637	\$575.3 Million
Condominiums	843	8.4%	\$406,949	\$343.1 Million
Townhomes	468	4.7%	\$490,287	\$229.5 Million
Multi-Family	400	4.0%	\$1,071,870	\$428.7 Million
Farm / Agriculture	186	1.9%	\$2,405,246	\$447.4 Million
Total / Average	9,999	100.0%	\$678,467	\$6.78 Billion

Commercial Takeaways

Single Family Dominance: Represents 63.8% of listing inventory and 70.1% of total transaction dollar value (\$4.76B).

High-Value Agricultural Outliers: Farm properties account for only 1.9% of units but command the highest average price (\$2.41M), driven by substantial acreage.

Density & Urban Alternatives: Condominiums and townhomes combine for 13.1% of inventory with entry-level price points (\$407k and \$490k).

Commercial Multi-Family Density: Multi-family averages \$1.07M, functioning as prime target assets for income investors.

Square Footage, Bathrooms, and Price per SqFt

Correlating physical space dimensions with market valuation efficiency

Property Type	Avg Square Feet	Avg Bathrooms	Avg Price / SqFt	Cost Efficiency
Multi-Family	4,161 sqft	4.64 baths	\$266.70	Highest Scale Discount
Farm / Ranch	2,758 sqft	2.46 baths	\$1,462.56	High Land Premium
Single family	2,202 sqft	2.28 baths	\$365.97	Core Benchmark
Townhomes	1,575 sqft	2.18 baths	\$316.13	Efficient Urban Layout
Condominiums	1,160 sqft	1.65 baths	\$356.21	Compact Footprint
Statewide Total	2,143 sqft	2.26 baths	\$365.74	Market Standard

Unit Economic Insights

Multi-Family Economies of Scale: At 4,161 average sqft and 4.64 baths, multi-family properties offer the lowest cost per square foot (\$266.70), reflecting investor focus on rentable yield over premium finishes.

Farm Outlier Dynamics: Farm price/sqft (\$1,462.56) is heavily skewed because listing valuations include raw acreage and outbuildings not captured in interior residential square footage.

Single Family vs Townhomes: Townhomes are priced at a \$50/sqft discount (\$316 vs \$366) compared to standalone single family homes, representing strong value for budget-conscious buyers.

Construction Eras and Historical Valuation Trends

Analyzing property counts and average list prices by decade of construction

Decade Era	Properties Listed	Average List Price	Pricing Premium / Trend
2020s Construction	1,400 units	\$814,399	Modern Build Premium
2010s Decade	548 units	\$869,580	High-End Quality Peak
2000s Expansion	1,218 units	\$788,815	Suburban Growth Cycle
1990s Decade	885 units	\$788,504	Stable Mature Housing
1970s Build Boom	955 units	\$667,023	Mid-Century Stock
1940s-1950s Post-War	896 units	\$535,240	Entry Level Affordability
Pre-1930s Historic	835 units	\$720,000+	Historic Architectural Value

Historical Era Analysis

New Construction Surge: Properties built in the 2020s represent the single largest modern era cohort (1,400 homes) with strong average valuation (\$814k).

2010s Premium: The 2010s decade exhibits the highest average list price (\$869,580), reflecting modern master-planned communities and high-end suburban developments.

Affordability Trough: 1940s and 1950s inventory presents the lowest pricing threshold (~\$535k), marking prime opportunities for renovation and value-add equity expansion.

Century-Old Longevity: Homes built between 1852 and 1920 maintain a price premium over mid-century homes due to urban core locations and preserved historic appeal.

Architectural Drivers: Stories and Garage Availability

Evaluating valuation premiums associated with vertical density and vehicle storage

Vertical Stories Pricing Progression (Q9)

- 1 Story: \$573,626 average list price (Baseline single-level living).
- 2 Stories: \$811,907 average list price (+41.5% premium over 1-story homes).
- 3 Stories: \$1,000,960 average list price (+23.3% premium over 2-story homes, crossing the million-dollar threshold).
- 4 Stories: \$1,384,321 average list price (High-end custom builds and urban luxury townhomes).
- 5 Stories: \$1,591,333 average list price (Luxury penthouse and multi-level complexes).
- Strategic Takeaway: Each additional vertical level correlates with an average valuation escalation of approximately \$200k to \$250k.

Garage Amenity Penetration (Q8)

- Statewide Adoption: 6,441 properties (64.4%) feature enclosed garages, while 3,558 properties (35.6%) lack garage access.
- Townhomes: Highest penetration at 86.5% (405 of 468 units have attached garages).
- Single Family: 83.7% penetration (5,343 of 6,383 units have garages). Homes lacking garages in this segment suffer severe marketing liquidity penalties.
- Condominiums: 60.1% penetration (507 of 843 units have dedicated garage space).
- Agricultural & Land: Only 17.7% of farms and 1.2% of raw land parcels record residential garage structures.
- Strategic Takeaway: Garage parking is a baseline consumer requirement for over 83% of suburban single-family buyers.

Interactive Excel Presentation Dashboard

A fully linked 10-chart executive dashboard interface with synchronized slicers

Visual Composition

- 10 Synchronized Charts: Bar, Pie, Doughnut, Line, and Area charts mapped to the 10 core analytical questions.
- Executive Banner: Prominent header 'OREGON REAL ESTATE PRESENTATION DASHBOARD' establishing corporate polish.
- Card Metrics: Top-level summary totals displaying portfolio volume (\$6.78B) and unit count (9,999).

Interactivity Engine

- Multi-Pivot Slicers: Filter connections enabling real-time drill-down by Property Type, Garage status, and Decade.
- Cross-Filtering: Adjusting any single slicer instantly cascades dynamic recalculations across all 10 pivot tables.
- Zero Latency: Optimized data tables structured for instant Excel recalculation without formula overhead.

Executive Experience

- Gridline Protection: Configured view panes and clean boundary framing eliminate spreadsheet visual clutter.
- Truncation Elimination: Column widths and chart labels formatted to prevent text clipping and numeric overflow.
- Presentation Ready: Designed for direct screen presentation in executive board meetings and investment committees.

Strategic Recommendations for Real Estate Operators

Commercial guidance synthesized from statewide valuation and inventory metrics

Recommendation 1: Value-Add Acquisition in Mid-Century Stock

Target post-war inventory constructed between 1940 and 1960 currently trading at the market affordability floor (\$535k average). By executing modernization capital expenditures (open-concept floor plans, primary suite additions, and garage additions), operators can capture the 40%+ valuation spread commanded by modern single-family stock.

Recommendation 2: Density Infill Development via Townhomes

With single-family prices averaging \$745k, townhomes at \$490k (\$316/sqft) provide the highest absorption velocity among first-time buyers. Developers should prioritize multi-story attached units featuring attached garages (commanding an 86.5% preference rate) to maximize land yield while maintaining sub-\$550k price points.

Recommendation 3: Commercial Multi-Family Portfolio Scaling

Multi-family properties offer the most favorable unit economics at \$266.70 per square foot across an average 4,161 square foot footprint. Commercial investors should capitalize on this scale discount to acquire high-density residential assets before rent-growth compression further narrows margin spreads.

PROJECT DELIVERABLES AND CONTACT

This analytical study forms part of the commercial intelligence and data analytics portfolio created by Mafeng Daniel.

Workbook Artifact: oregon_real_estate_2026_ultimate.xlsx
Statewide Listings Analyzed: 9,999 properties
Total Market Volume: \$6,783,995,305

GitHub Portfolio: Available in primary portfolio repository
Direct Email: mafengdaniel15@gmail.com